



Assessing changes to nutrient density and availability following separation, drying, and pyrolysis of manure solids

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ABSTRACT

Manure nutrients are beneficial when land applied but contribute to environmental degradation when lost as runoff or emissions. Applying manure nutrients to better meet agronomic crop needs has potential to reduce losses. However, transporting manure nutrients to nutrient deficient fields is commonly cost prohibitive. Densifying manure nutrients has the potential to reduce costs. Manure processing of separated manure solids via drying at 50 °C for 48 h reduced the mass by 62.8 %, where reduction by pyrolysis reduced the mass by 81.8 % and 83.4 % at 350 and 500 °C, respectively. Drying and pyrolysis retained and densified the manure phosphorus. Drying retained the nitrogen within the solids, but 37 to 47 % of nitrogen was lost through pyrolysis. Immediately after processing, available phosphorus decreased with increasing pyrolysis temperature. Incubation of manure solids and processed manures shows that processing decreased available phosphorus through the first 28 days, but following that initial period reached similar phosphorus availability.

1. Introduction

Over many decades manure has been land applied to return organic matter to the soil and provide essential nutrients, such as nitrogen (N) and phosphorus (P), to improve soil fertility. However, overapplication of manure nutrients that exceed crop nutrient uptake can result in the build-up of nutrients in the soil, making them more susceptible to environmental losses and result in poor manure nutrient use efficiency. Historically, over application of manure nutrients was due to the imbalance of N and P in livestock manure with respect to crop needs, resulting in overapplication of P to meet N needs leading to soil buildup, also known as legacy P (Sharara et al., 2022). Further, increases in animal density has increased imported feed thus increasing nutrients to a given land base. The resulting nutrient imbalance, particularly for P surpluses, requires larger land bases than commonly available near the livestock farm to distribute manure at agronomic rates (Spiegel et al., 2020). Export of manure nutrients to nutrient deficient locations could serve to improve nutrient balances, allow agronomic application rates, and thus reduce losses. However, densification of manure nutrients is required to make transport more feasible as the value of the nutrients alone is not sufficient to support the cost of transport beyond a few miles.

Manure processing can be used to densify manure nutrients. Solid liquid separation is a common manure processing technology adopted at livestock facilities (Aguirre-Villegas and Larson, 2017). Solid liquid separation systems, such as inclined screens, screw press or centrifuge, can separate 25–80 % of the manure P into the solids fraction (Aguirre-Villegas et al., 2019; Jørgensen and Jensen, 2009; Peters et al., 2011). However, despite the densification of manure P in the solids, studies identified separation alone to be insufficient in many circumstances to reduce the economic burdens of transport (Nolan et al., 2012; Pellervo et al., 2013). Thus, producers are looking for additional methods to further densify manure nutrients to increase the economically viable transport distance.

Pyrolysis thermochemically converts feedstocks, to produce syngas, bio-oils and biochar. The syngas and bio-oil fractions are primarily used for energy (Spokas et al., 2012), while biochar has been used to sequester carbon, improve soil fertility, and reduce nutrient leaching (Lehmann and Joseph, 2015). Livestock manure has been used as a feedstock for biochar production in several studies and proven effective at significantly reducing the mass of livestock manure and densifying P (Cantrell et al., 2012; Cao and Harris, 2010; Cely et al., 2015; Uzoma et al., 2011). Cantrell et al. (2012) produced biochar from swine separated-solids, paved-feedlot manure, dairy manure, poultry litter,

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and turkey litter at 350 and 700 °C, finding high recovery of P in the biochar. Other studies have found similar results when producing dairy manures into biochar at temperatures below 500 °C (Cao and Harris, 2010). While the process of densifying manure P through pyrolysis has been evaluated, few studies have investigated P fertilizer potential of manure derived biochar. Jin et al. (2016) evaluated swine manure biochar produced at 400 °C in a 98-day incubation study when amending biochar to soils at 0.5 and 1.5 % and found soil P increased by 18 to 34 % at the end of the study. Liang et al. (2014) converted dairy manure into biochar at 450 °C. Raw manure or biochar was then added to soil at 2.5 to 5.0 % dry weight, which found a slow release of available P from biochar products compared to manure. Subedi et al. (2016) applied manure based biochars to a controlled pot study and found that ryegrass was able to uptake nutrients from the biochar demonstrating its ability to serve as an organic fertilizer. While the before mentioned studies are useful in assessing biochar as a fertilizer, these studies did not compare the original manure to the biochar at agronomic rates of P making assessments to existing systems difficult. A 14-day study conducted by Bruun et al. (2017) found pyrolysis of digested manure and food waste had little impact to P speciation at temperatures below 600 °C, finding the immediate P availability decreased but then increased to the same level as the digested separated solids after 14 days of incubation. This provides more information but does not consider longer term effects with a very short incubation period. This study aims to evaluate the densification of manure P following separation, drying, and pyrolysis and the P availability in soils throughout a 182-day incubation study.

2. Methods

2.1. Soil samples and biochar preparation

A loam soil (20 % sand, 54 % silt, and 26 % clay) was collected from an agricultural site currently in a corn, soybean, alfalfa rotation. Soil texture was assessed at the University of Wisconsin Soil and Forage Laboratory. To provide additional information on impacts of sandy soil, an engineered sandy loam soil (55 % sand, 30 % silt, and 15 % clay) was prepared by blending the collected loam soil and sand from a local landscaping company. Soil was characterized for pH (1:1 soil to solution ratio), available P (Bray-1P method), TN (total Kjeldahl nitrogen), nitrate (KCl method) and ammonium (KCl method) (see the Supplementary materials).

Manure separated solids were collected from a dairy facility in WI, USA. Manure produced from the 2500 head dairy on a forage feed diet is processed through a plug flow anaerobic digester and a screw press manure separator. Fresh separated manure solids were collected for the study during separation. Manure solid samples were transported back to laboratory and a fraction was immediately placed in a -20 °C freezer until needed for incubation while the rest was immediately dried in a force air drying oven at 50 °C for 72 h. Biochar was produced from the dried manure solids in a modified Fischer Scientific Lindberg/Blue M Moldatherm box furnace (Thermo Fisher Scientific, Waltham MA, United States) fitted with an Omega CN9600 SERIES Autotune Temperature Controller (Omega Engineering Inc. Norwalk, CT, United States). Manure solids were placed in a steel cylinder inside the furnace which was subjected to argon gas at 1 L min⁻¹ to maintain anaerobic conditions. The heating rate was set to 10 °C min⁻¹ and biochar was produced at two different temperatures, 350 and 500 °C to provide data for a low and high temperature biochar to evaluate differences, and temperature held for 1 h. Initial and final masses before drying and pyrolysis were recorded to evaluate mass and nutrient reductions.

Separated manure solids (wet), dry manure solids, and pyrolyzed manure solids were analyzed for total solids (TS), total nitrogen (TN), and total phosphorus (TP) following recommended methods for manure analysis (Peters et al., 2003). Total solids (or dry matter content) were determined gravimetrically and dried in a 110 °C oven for 24 h. Manure

and biochar TN and TP was determined using a Kjeldahl digestion (copper catalyst digested at 380 °C) and analyzed on a Seal Analytical AQ2 (SEAL Analytical, Norderstedt, Germany) following EPA-136 and EPA-135 methodology, respectively. Manure and biochar pH were determined following IIB (2015) recommended method using a 1:20 biochar/manure to deionized water ratio (w/v) and shaking for 90 min and measuring equilibrium pH with a HQ440d Benchtop Multi Meter pH probe (Hach Company, Loveland, CO, United States). Characteristic data for manure and biochar products can be found in the Supplemental materials.

2.2. Soil incubation

Incubation studies were carried out in 1 L polyethylene bottles. Each bottle contained 800 g of dried soil, which was brought up to 40 % of its water-holding capacity and pre-incubated at 25 °C for 7 days to revive soil microbial activity. Sets of triplicates from each soil texture (loam and sandy loam) were set aside as control samples, and the remaining soils received separated manure solids (wet), dry manure solids, or biochars. Treatments were added to the bottles at two agronomic rates with respect to P content in triplicate. The first application rate was 34.5 mg P kg⁻¹, which is equivalent to P recommendation for optimal corn silage yield (79.4 kg P₂O₅ ac⁻¹) following A2809 (Laboski et al., 2006), and the second application rate was 172.4 mg P kg⁻¹ (or 396 kg P₂O₅ ac⁻¹) to evaluate higher application rates. For the lower application rate these applications were equivalent to 1.2, 0.5, 0.2, and 0.2 % (w/w) mass amendment for wet MS, dry MS, BC350, and BC500, respectively. For the higher application rate these applications were equivalent to 6.0 %, 2.4 %, 1.2 %, and 1.1 % (w/w) mass amendment for wet MS, dry MS, BC350, and BC500, respectively.

Bottles were covered and kept aerated in incubator at 25 °C for 182 days. Samples were collected on day 0, 3, 7, 14, 21, 28, 42, 56, 98, 112, 126, 140, 154, and 182 during the incubation to evaluate the release of available P. During the soil incubation the soil moisture content was kept constant at 40 % of the soils water-holding capacity, based on methods from past incubation studies (Jin et al., 2016; Liang et al., 2014). Soil moisture content was recorded weekly and deionized water added to incubation bottles as needed. It is important to note that a gap in sampling between days 28, 56 and 98 was due to COVID 19 restrictions at the laboratory facility putting a temporary hold on sample collection and moisture content adjustments.

2.3. Soil nutrient analysis

Soil incubation samples were dried in a forced air-drying oven at 50 °C for 72 h and then mechanical ground to pass through a 2 mm sieve. Soil pH was measured using a 1:1 (w/v) soil and water suspension and measured with a HQ440d Benchtop Multi Meter pH probe (Hach Company, Loveland, CO, United States). Soil TN and TP was determined using a Kjeldahl digestion (copper catalyst digested at 380 °C) and analyzed on a Seal Analytical AQ2 (SEAL Analytical, Norderstedt, Germany) following EPA-136 and EPA-135 methodology, respectively. Available P was assessed using the Bray 1P method. Extractant (0.025 N HCl, 0.03 N NH₄F) was added to soil at a 1:1 (w/v) ratio, placed on an orbital shaker at 200 rpm for 30 min, then filtered. Filtrate was analyzed for orthophosphate on the AQ2 following EPA-146 molybdenum blue method. Inorganic N was determined using a 2 N KCl extraction (1:1 (w/v) ratio) and analyzed for NO₂+NO₃-N and NH₄⁺-N following EPA-141 and EPA-153, respectively.

2.4. Data analysis

Statistical analysis was conducted in RStudio (RStudio Team, 2015). Manure solids, dry manure soils, and biochars TS, TN, and TP composition and mass, N, and P recoveries were compared using analysis of variance (ANOVA) with a significance level of 0.05. Further analysis was

Table 1

Composition of wet dairy manure solids, dry dairy manure solids, and biochars and treatment recoveries. Values represent averages of values (standard deviation), different letters represent significant difference ($p = 0.05$) between treatments.

Treatment	Composition				Constituent recovery**		
	TS (%)	TN (g/kg)	TP (g/kg)	Available P* (mg/kg)	Mass (%)	TN (%)	TP (%)
Wet MS	28.5 (0.9) ^a	8.3 (0.1) ^a	2.9 (0.1) ^a	415 (37) ^b	—	—	—
Dry MS	93.6 (2.0) ^b	22.2 (1.3) ^b	7.5 (0.6) ^b	482 (16) ^a	37.2 (0.7) ^a	99.4 (0.8) ^a	99.7 (0.6) ^a
BC350	97.6 (0.2) ^c	28.6 (1.3) ^c	15.3 (1.6) ^c	327 (11) ^c	18.2 (0.2) ^b	62.4 (0.2) ^b	98.9 (0.3) ^a
BC500	99.6 (0.1) ^c	28.4 (1.1) ^c	16.7 (0.6) ^c	200 (18) ^d	16.6 (0.2) ^c	56.7 (0.3) ^c	98.8 (0.5) ^a

* Determined via Bray-P method.

** Constituent recovery is the mass (or mass of nutrient) after processing divided by mass before processing.

conducted using a Tukey honestly significant difference (HSD) test to compare composition and recovery. Nutrient content for incubation samples were compared using ANOVA with a significance level of 0.05 and treatments compared using pairwise comparison Tukey method with *emmeans* package.

3. Results and discussion

3.1. Densification of manure phosphorus

Mass reduction and content of P and N were evaluated in manure solids (Wet MS), dried manure solids (Dry MS), and pyrolyzed manure solids (BC350 and BC500) (Table 1). Drying of manure solids significantly reduced the mass by 62.8 %, where reduction by pyrolysis reduced the mass from the original manure by 81.8 % at 350 °C and 83.4 % at 500 °C. The large mass reduction was a result of the drying with further pyrolysis contributing additional mass reduction.

Recovery of P in the dried separated manure solids and the manure

based biochars was above 98 % for all processing treatments, which was expected as temperatures did not exceed the 700 °C needed for phosphorous volatilization (DeLuca et al., 2015). The increase in P following drying is due to reduction in moisture content of the manure and following pyrolysis due the thermochemically breakdown of organic material into syngas and bio-oil, further reducing the materials mass. Concentration of P in the manure derived biochar at 350 and 500 °C was 5.3 and 5.8 time higher than original manure solids product, respectively. The concentration of manure P into biochar was similar to past studies (Liang et al., 2014; Wang et al., 2012). Available P, determined by Bray-P methodology, decreased following pyrolysis (Table 1), similar to past observations of water soluble P availability (Cantrell et al., 2012; Liang et al., 2014; Uchimiya et al., 2015) and decreased with increasing temperature as observed in Subedi et al. (2016). Reduction in available P is likely due to formations of more stable P forms on the surface of the biochar. Uchimiya et al. (2015) and Liang et al. (2014) both found evidence of stabilization of soluble P into crystallized struvite or whitlockite on biochar following pyrolysis. However, drying of the solids

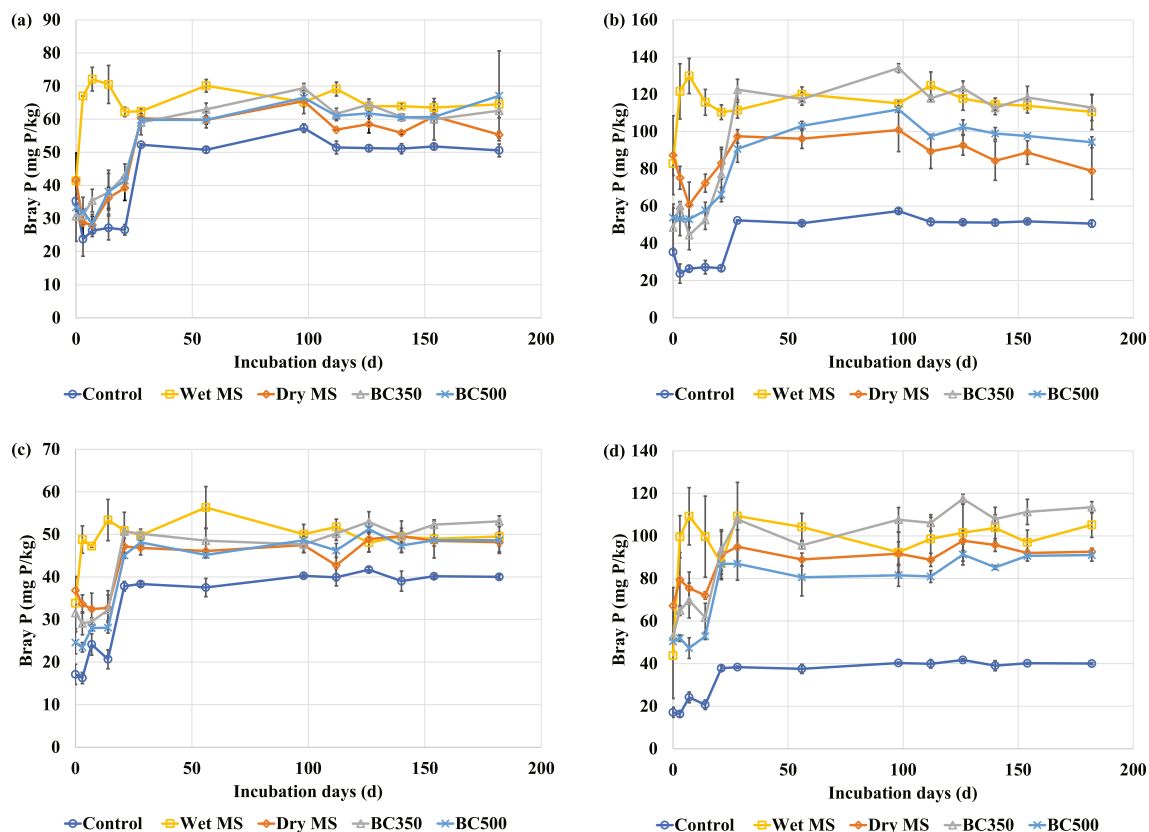


Fig. 1. Soil available P concentrations (measured using Bray P methods) throughout the incubation period for (a) loam application rate of 79.4 kg P₂O₅ ac⁻¹, (b) loam application rate of 396 kg P₂O₅ ac⁻¹, (c) sandy loam application rate of 79.4 kg P₂O₅ ac⁻¹, (d) sandy loam application rate of 396 kg P₂O₅ ac⁻¹. Error bars represent standard deviation between triplicates. Sample point treatment averages, standard deviations, and letters of significant difference can be found in the Supplemental material.

slightly increased the available P and both the original manure and the dried separated solids had a greater P availability than the biochars.

Drying the separated manure solids resulted in almost no loss of N at only 0.6 %. The low percent loss of N is likely due to low temperature drying and because separated manure solids have low ammoniacal N as most of the N is in organic form (Aguirre-Villegas et al., 2019). Pyrolysis resulted in an increase in N loss of 37.6 and 43.3 % at 350 and 500 °C, respectively. Losses during pyrolysis were likely due to volatilization and emissions of ammonia and N₂ (Cantrell et al., 2012; Novak et al., 2009). While there were increased losses of N, the TN content increased with pyrolysis at 350 and 500 °C, which were both 3.4 times higher than the original manure solids feedstock due to the mass reduction. The composition change is different than that of Cantrell et al. (2012), which only observed a 0.9 time increase following biochar produced from dairy manure at 350 °C.

3.2. Nutrient availability from manure-derived products during incubation

For all P application treatments, the applied products increased the P availability in the soil when compared to the control (Fig. 1). The separated manure solids (Wet MS) for all treatments started with a high available P content and remained high throughout the trial. This was similar to results observed in Bruun et al. (2017) when comparing digested solids to biochars at three different temperatures over fourteen days. The Dry MS and both biochars (BC350 and BC500) followed similar trends in terms of P availability, starting low and increasing over time until having similar concentrations to Wet MS after 28 days. The lower starting P availability in the soil is likely due to P speciation following processing treatment, as both biochars had a significantly lower available P content following pyrolysis compared to separated manure solids (Table 1). The Wet MS had a higher P availability than the dry manure solids or biochars in the first 21 to 28 days depending upon the treatment (application rate and soil type) (Fig. 1). From this time forward, the P availability remained consistent throughout the trial for all treatments, similar to a shorter study by Bruun et al. (2017). At the end of the 182-day incubation study, for lower application rate of 79.4 kg P₂O₅ ac⁻¹ there was no significant difference between the four treatments in both soils in regard to available P (Fig. 1a and c). However, for the higher P application rate of 396 kg P₂O₅ ac⁻¹, BC350 was significantly different than BC500 and Dry MS. The lower temperature biochar having higher soil P availability is similar to past studies (Bruun et al., 2017), and is likely a result of differences in P speciation at different pyrolysis temperatures.

As the N was applied at different rates (due to differences in N:P ratios in each fertilizer product), direct treatment comparisons are not possible. However, all treatments received some additional N when compared to the control, but it should be noted there was little difference in ammoniacal N or NO₂+NO₃-N concentrations throughout the trial (see the Supplementary materials). The ammoniacal N dropped for all treatments within the first seven days for both the loam and sandy loam soils. Following that time, all treatment ammoniacal N concentrations were approximately 5 mg N kg⁻¹ for all treatments and soil types indicating no significant difference from the control. The NO₂+NO₃-N slowly increased throughout the trial for all treatments with similar trends (see the Supplementary materials). However, the treatments provided no more NO₂+NO₃-N than the control that did not receive fertilizer. Therefore, the applied processed fertilizers have provided no additional N availability in the first 182 days of incubation.

4. Conclusion

Pyrolyzing manure solids densified separated manure solids up to 84 % and concentrated manure P, but decreased P availability by 22 to 52 %. However, during the incubation trials while P availability is reduced initially for pyrolyzed manure solids compared to the wet manure solids,

after 28 days similar available P concentrations were achieved. Pyrolyzing manure solids did result in a 38 to 43 % loss of N with increasing temperatures, as expected. When incubated the ammoniacal N and NO₂+NO₃-N followed similar trends to the control indicating little N fertilizer from the biochar over the 182 days of the incubation trial.

CRediT authorship contribution statement

J.R. Sanford contributed to the conceptualization, methodology, conceptualization, laboratory analysis, completed the formal analysis and writing of the original draft preparation. D. Johnstone contributed to laboratory analysis and data analysis. R.A. Larson contributed to the conceptualization, methodology, writing—review and editing, supervision, project administration, and funding acquisition. All authors read and approved the final manuscript.

Declaration of competing interest

Rebecca Larson reports financial support was provided by U.S. Department of Agriculture.

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Data availability statement

All data generated or analyzed during this study are included in this published article [and its Supplementary materials files].

Appendix A. Supplementary data

Appendix A includes Supplemental material including additional characterization of soil, manure, and biochar, tables of significance for phosphorus incubation data, and soil ammoniacal nitrogen and nitrate and nitrite data. Supplementary data to this article can be found online at <https://doi.org/10.1016/j.biteb.2022.101169>.

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